

Auteur: Michael Livchits
Studierichting: Informatica
Oefeningenreeks 7E nr. 1.9

$$f(x, y) = 10 + 12x - 12y - 3x^2 - 2y^2$$

Eerste orde:

$$\frac{\partial f}{\partial x} = 12 - 6x \Rightarrow x = 2$$

$$\frac{\partial f}{\partial y} = -12 - 4y \Rightarrow y = -3$$

Kritiek punt: $(2, -3)$

Hessian matrix:

$$\Rightarrow D^2 f(x, y, z, t) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} = -6 & \frac{\partial^2 f}{\partial y \partial x} = 0 \\ \frac{\partial^2 f}{\partial x \partial y} = 0 & \frac{\partial^2 f}{\partial y^2} = -4 \end{pmatrix}$$

$$\det H(x, y) = 24$$

$$H(2, -3) = 24 > 0 \text{ en } \frac{\partial^2 f}{\partial x^2} = -6 < 0 \Rightarrow \text{bereikt een maximum}$$

References